

MATH 301: Homework 5

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Section 3.4 : Use method of *Example 3.4.3(b)* to show $0 < c < 1 \implies \lim c^{\frac{1}{n}} = 1$

Example (b): Proving $\lim(c^{1/n}) = 1$ for $c > 1$

The Goal: Prove that the n -th root of a number greater than 1 gets closer and closer to 1 as n gets larger. Let our sequence be $z_n = c^{1/n}$.

Step 1: Show the sequence is decreasing

The text asks "(Why?)" here. Here is the reason: Since $c > 1$, taking a smaller and smaller fractional power of c yields a smaller and smaller result. Because $\frac{1}{n+1} < \frac{1}{n}$, it follows that:

$$\begin{aligned}c^{1/(n+1)} &< c^{1/n} \\ z_{n+1} &< z_n\end{aligned}$$

So, the sequence is decreasing.

Step 2: Show the sequence is bounded

Because $c > 1$, any positive root of c will still be greater than 1.

$$z_n > 1$$

The sequence is bounded below by 1.

Step 3: Conclude it converges

Just like in example (a), we have a decreasing sequence that is bounded below. By the Monotone Convergence Theorem, it converges. Let's call the limit z :

$$z = \lim(z_n)$$

Step 4: Find the limit using the same trick

Let's look at the subsequence of even terms, z_{2n} . Because the main sequence converges to z , this subsequence also converges to z . Let's look at the algebra:

$$z_{2n} = c^{1/(2n)} = (c^{1/n})^{1/2} = \sqrt{z_n}$$

If we square both sides to get rid of the fraction, we get:

$$(z_{2n})^2 = z_n$$

Now, take the limit of both sides as $n \rightarrow \infty$:

$$z^2 = z$$

Step 5: Solve for z

Again, the only solutions to $z^2 = z$ are $z = 0$ and $z = 1$. Since we established in Step 2 that every term in our sequence is strictly greater than 1 ($z_n > 1$), the limit cannot possibly be 0. Therefore, the limit must be 1.

$$\lim(c^{1/n}) = 1$$

$$c^{\frac{1}{n+1}} < c^{\frac{1}{n}} \implies c^{\frac{1}{n}} \text{ is decreasing}$$